Lab Report No. 02

Experiment Name

Drawing Multiple Colored Geometric Objects Using OpenGL.

Objective

To create a 600×600 OpenGL window and draw four non-overlapping colored objects: a square, a triangle, a rectangle, and a pentagon at different positions on the screen.

Theory/Introduction

OpenGL is a graphics library used to create and display 2D and 3D graphics. Different geometric shapes can be drawn using OpenGL primitives such as triangles, quadrilaterals, and polygons.

In this experiment, four different colored objects are drawn in separate corners of a 600×600 window:

- A small green square in the bottom-left corner.
- A large purple triangle in the top-right corner.
- A large rectangle made of two triangles with different colors in the top-left corner.
- A small orange pentagon in the bottom-right corner.

Algorithm/Procedure

1. Create a 600×600 OpenGL window.
2. Set the background color.
3. Draw a small green square in the bottom-left corner.
4. Draw a large purple triangle in the top-right corner.
5. Draw a rectangle in the top-left corner using two triangles of different colors.
6. Draw a small orange pentagon in the bottom-right corner.
7. Display all objects in the OpenGL window.

Source Code

```
from OpenGL.GL import *
from OpenGL.GLUT import *
from OpenGL.GLU import *

def display():
    glClear(GL_COLOR_BUFFER_BIT)
```

```
    glLoadIdentity()
    glOrtho(0, 600, 0, 600, -1, 1)

    # Small Green Square (Bottom-Left)
    glColor3f(0, 1, 0)
    glBegin(GL_QUADS)
    glVertex2f(50, 50)
    glVertex2f(150, 50)
    glVertex2f(150, 150)
    glVertex2f(50, 150)
    glEnd()

    # Large Purple Triangle (Top-Right)
    glColor3f(0.6, 0.0, 0.8)
    glBegin(GL_TRIANGLES)
    glVertex2f(400, 400)
    glVertex2f(550, 400)
    glVertex2f(475, 550)
    glEnd()

    # Rectangle using Two Colored Triangles (Top-Left)

    # Triangle 1 (Blue)
    glColor3f(0, 0, 1)
    glBegin(GL_TRIANGLES)
    glVertex2f(50, 400)
    glVertex2f(250, 400)
    glVertex2f(250, 550)
    glEnd()

    # Triangle 2 (Red)
    glColor3f(1, 0, 0)
    glBegin(GL_TRIANGLES)
    glVertex2f(50, 400)
    glVertex2f(50, 550)
    glVertex2f(250, 550)
    glEnd()

    # Small Orange Pentagon (Bottom-Right)
    glColor3f(1.0, 0.5, 0.0)
    glBegin(GL_POLYGON)
    glVertex2f(450, 80)
    glVertex2f(500, 120)
    glVertex2f(480, 180)
    glVertex2f(420, 180)
    glVertex2f(400, 120)
    glEnd()

    glutSwapBuffers()

glutInit()
```

```
glutInitDisplayMode(GLUT_RGBA | GLUT_DOUBLE)
glutInitWindowSize(600, 600)

glutCreateWindow(b"Marajul Islam - 28")

glClearColor(0.0, 0.0, 0.0, 1.0)

glutDisplayFunc(display)
glutMainLoop()
```

Sample Input

No user input is required.

Objects to be drawn:

- Green Square
- Purple Triangle
- Rectangle using Two Colored Triangles
- Orange Pentagon

Sample Output

A 600×600 OpenGL window displaying:

- Green square at the bottom-left corner
- Purple triangle at the top-right corner
- Rectangle (formed by blue and red triangles) at the top-left corner
- Orange pentagon at the bottom-right corner

Result

The OpenGL program successfully displayed four non-overlapping colored geometric objects in their respective positions within the 600×600 window.

Conclusion

The experiment was completed successfully. Different OpenGL primitives were used to draw multiple geometric objects with different colors and positions. The required output was achieved correctly.